



Green Computer Vision: Comparing Accuracy and Efficiency in Deep Image Classification

Main Topic: Energy-Efficient Deep Learning for Image Classification



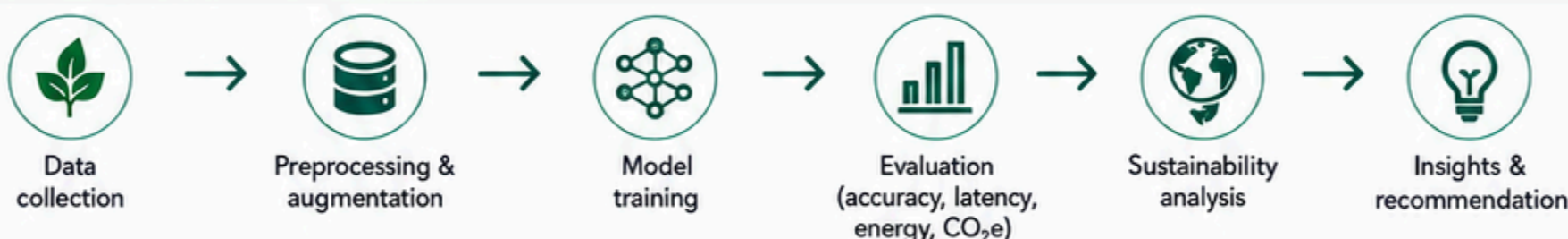
1. Motivation

- Deep learning drives powerful image classification but can be energy-intensive and costly to deploy.
- We compare lightweight and heavier CNN architectures to evaluate accuracy, latency, energy use, and model efficiency.
- Goal: Identify models that deliver high performance with low resource consumption for sustainable, real-world applications.

2. Datasets

- | PlantDoc Dataset | TrashNet Dataset |
|--|---|
| Plant disease classification | Waste classification |
| ~25,000 images | ~2,527 images |
| 13 plant species | 6 waste categories (glass, metal, paper, plastic, cardboard, trash) |
| 26 disease classes (RGB, field captured) | (RGB) |

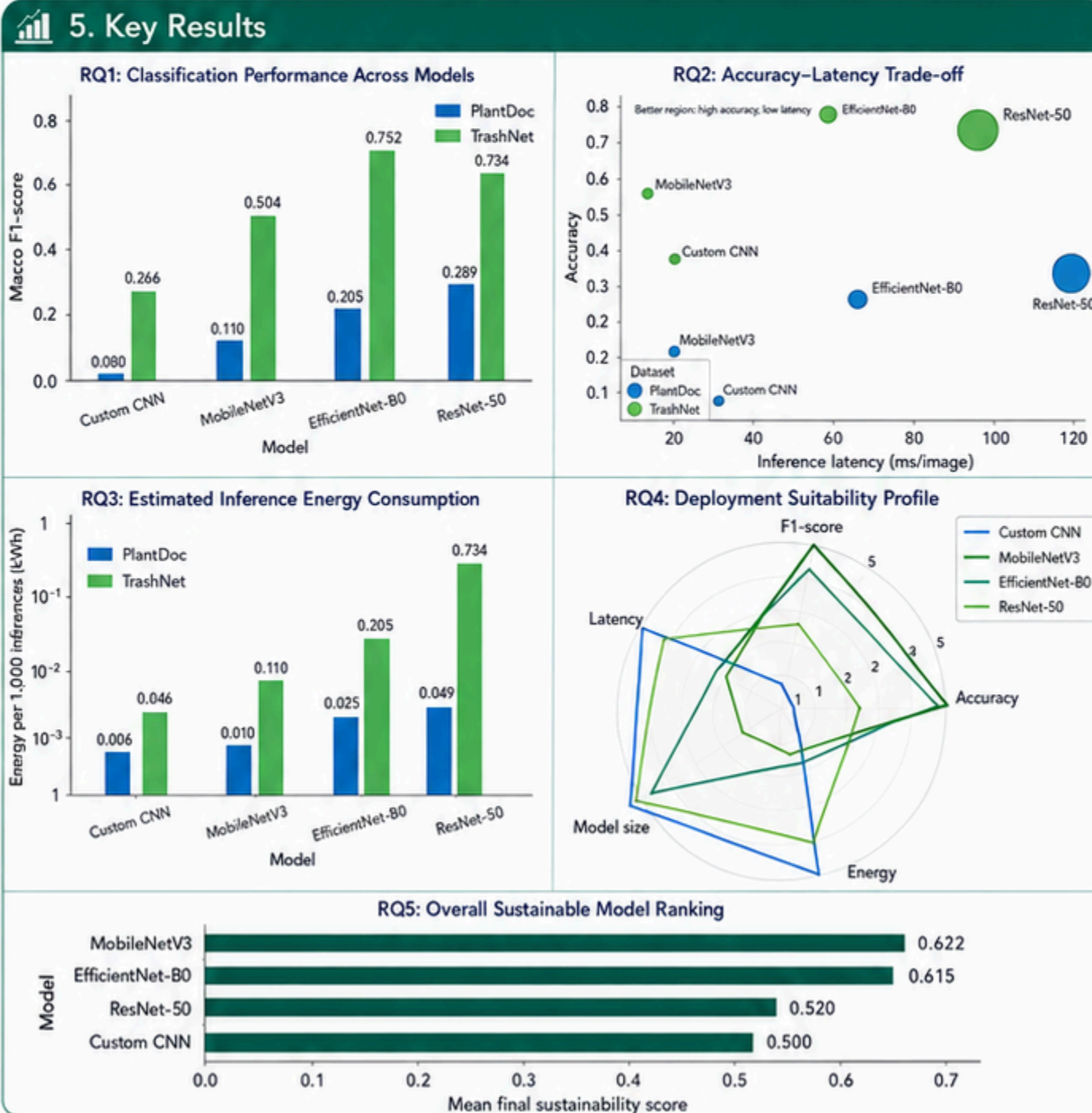
3. Methodology Pipeline



4. Research Questions

- RQ1: How do CNN architectures compare in classification performance on PlantDoc and TrashNet?
- RQ2: What are the trade-offs between accuracy, latency, and efficiency?
- RQ3: How much inference energy is consumed by different models?
- RQ4: Which models are most suitable for deployment on resource-constrained devices?
- RQ5: Which model gives the best overall sustainability ranking?

5. Key Results



6. Contributions

- Comprehensive comparison of four CNN architectures on two sustainability-focused datasets: PlantDoc and TrashNet.
- Joint evaluation across accuracy, latency, inference energy, and deployment suitability.
- Multi-criteria sustainability ranking to guide model selection for real-world deployment.
- Energy and carbon footprint analysis for inference at scale.
- Reproducible workflow with code, models, and evaluation pipeline.



Our results show that MobileNetV3 and EfficientNet-B0 offer the best balance of accuracy and efficiency across both PlantDoc and TrashNet. They achieve strong performance while maintaining moderate latency and energy consumption. ResNet-50 delivers the highest accuracy but at a substantially higher cost in latency, energy, and model size. Custom CNN is lightweight and fast but lags significantly in accuracy.

Takeaways

- Choose MobileNetV3 or EfficientNet-B0 for sustainable, real-time applications.
- ResNet-50 is suitable when maximum accuracy is critical and resources are ample.
- Lightweight models enable deployment on edge and resource-constrained devices.
- Energy-aware model selection reduces operational cost and carbon footprint.
- Sustainability metrics are essential for responsible AI deployment.

